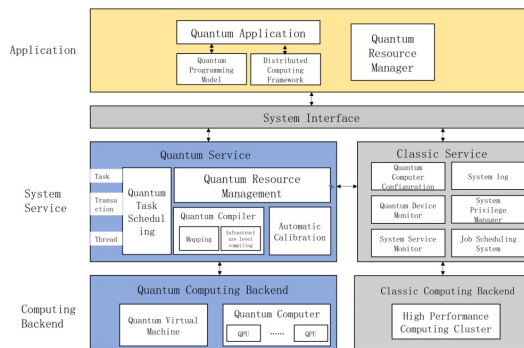

Origin Pilot The First Quantum Operating System

Caden W. Clements
CSC 9010: Program Quantum
Computers
Final Project

Introduction



- Launched in 2021 by Origin Quantum
- China's first Quantum Operating system
 - Now: Central command hub for China's quantum computing Operations
- Hardware Integration:
 - Superconducting, Trapped ions, Semiconductors, Neutral atoms, and Photonics.
- Supporting different computing backends:
 - quantum processors, quantum virtual machines and high-performance computing clusters

Overview

Launched in 2021 by Origin Quantum

Known as China's first quantum computer operating system

Acts as a central command hub for quantum computing operations and scheduling

Hardware Integration (Downstream)

Provides unified access to multiple quantum hardware types:

Superconducting circuits

Trapped ions

Semiconductors

Neutral atoms

Photonics

Contributions of Origin Pilot

Calibrate one or many qubits without interrupting other circuits

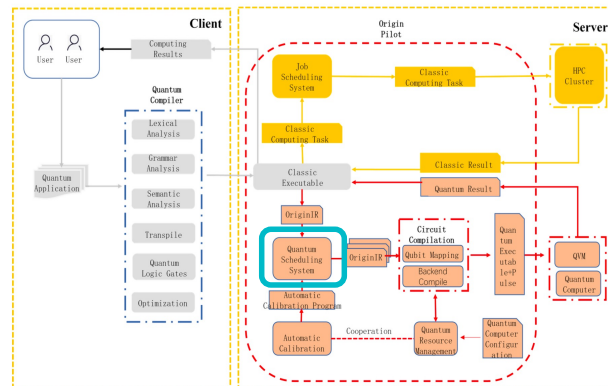
Ensuring it is in the correct starting state – CHECK FOR THIS

can allocate quantum resources for quantum circuits according to the state of the quantum resources and the requirements of quantum circuits

Reduce the decoherent noise with dynamic decoupling in qubits

Task Scheduling

- Multiple tasks on multiple processors.
- Elements of a quantum task
 - # of qubits
 - a program's intermediate representation
 - Processor's ID
 - Type of task: General or a calibration task
 - Priority of the task
- Highest Response Ratio Next – HRRN
 - $R_p = \frac{T_{waitingtime} + T_{runtime}}{T_{runtime}} = \frac{T_{response\,time}}{T_{runtime}}$



Scheduling multiple quantum tasks on multiple quantum processors

Quantum task elements

- (1) The number of qubits required for a quantum task;
- (2) Quantum program's intermediate representation;
- (3) Quantum processor's ID;
- (4) Type of a quantum task: a quantum task can be a general quantum task or automatic calibration task;
- (5) Priority: The priority of a quantum task.

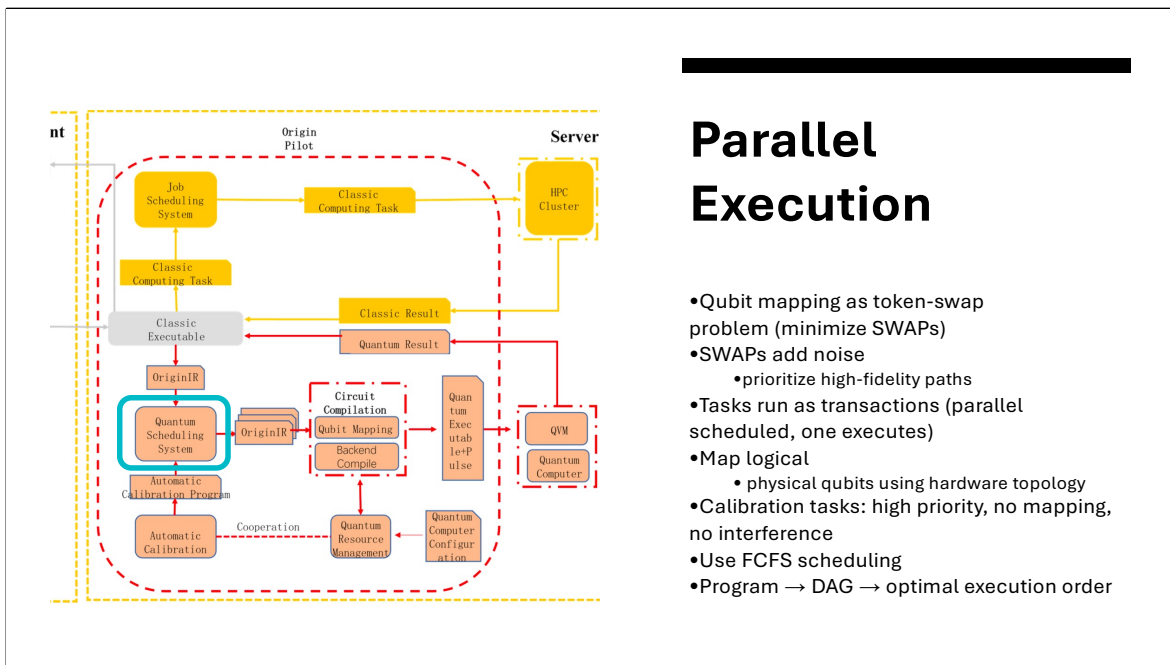
Highest Response Ratio Next – HRRN

$$R_p = \frac{T_{waitingtime} + T_{runtime}}{T_{runtime}} = \frac{T_{response\,time}}{T_{runtime}}$$

Some of the classical algorithms include

First come first serve

Shortest Job First



Qubit mapping is treated as a token-swap problem, where minimizing SWAP gates is critical due to added error and exponential complexity.

SWAP gates introduce additional noise, so mapping aims to both reduce swaps and choose routes with the highest fidelity.

Quantum tasks are scheduled and executed in parallel using transactions, but only one transaction runs at a time.

Before execution, logical qubits must be mapped to physical qubits based on the processor's topology.

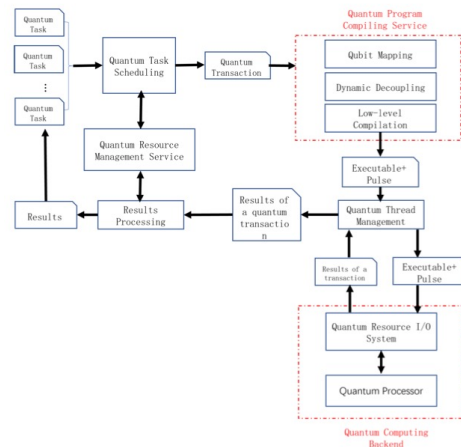
Automatic calibration tasks are given higher priority and do not require qubit mapping, preventing interference with other tasks.

Calibration tasks follow a First-Come-First-Served (FCFS) scheduling strategy.

The mapping process involves converting the quantum program into a Directed Acyclic Graph (DAG) and identifying optimal subgraph sequences.

Automatic Calibration

- Coherence time can help derive coupling strength between the system and quantum environment
- Qubit quality is affected by multiple factors
 - Automatic calibration is required
- Calibration Steps
 - Performance checking
 - Error adjustment
- Processors splits tasks in different regions
 - Improving fidelity



Automatic Calibration

Qubits' coherence time can be used to describe the coupling strength between a quantum system and the environment.

The longer they hold the state the better it gets

Magnetic can disturb the environment

Qubit quality depends on multiple factors, so continuous calibration is required.

Calibration has two steps: check performance → adjust errors.

Uses a POMDP model to automate calibration decisions.

A **Partially Observable Markov Decision Process (POMDP)** models calibration by letting an agent use noisy measurement observations to infer hidden hardware states and choose actions that maximize fidelity while minimizing calibration cost over time.

Processor is split into execution and calibration regions.

Tasks run on the best-performing region, improving fidelity.

Conclusions

- Still in early stages
- Core features accomplished
 - task scheduling, parallel execution, and calibration.
- Resource Management
 - Another accomplishment but not in this presentation
- Future work focus on bridges between quantum and Classical
- Origin Pilot is available for download!
 - open-source ecosystem development



Conclusions

Quantum operating systems are still early-stage research.

Origin Pilot provides core features: task scheduling, resource management, parallel execution, and calibration.

Future work includes distributed quantum computing and hybrid classical-quantum systems.

Plans to release the source code publicly for non-commercial use.

References

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